Traditional irrigation is more than just a production strategy

R code

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Contents

```r
# PRELIMINARY FUNCTIONS ########################################################

library(sensobol)
sensobol::load_packages(c("data.table", "ggplot2", "sf", "rnaturalearth",
                          "rnaturalearthdata", "ggrepel", "here", "readxl",
                          "ggwordcloud", "cowplot", "benchmarkme", "treemapify"))

# Source all .R files in the "functions" folder -------------------------------

r_functions <- list.files(path = here("functions"),
                          pattern = "\\.R$", full.names = TRUE)
invisible(lapply(r_functions, source))
```

1 Interview locations

```
# INTERVIEW LOCATIONS ##########################################################

locations <- data.table(
  place = c("Dúrcal", "Huétor-Tajar", "Murcia", "Ricote", "Sudanell", "Arriate", "Oria"),
  lon   = c(-3.5694,  -4.0439,        -1.1307,  -1.4361,   0.6011,    -5.0606,   -2.2731),
  lat   = c(37.0022,   37.1964,        37.9922,  38.0714,  41.5261,    36.7914,   37.5225)
)
```

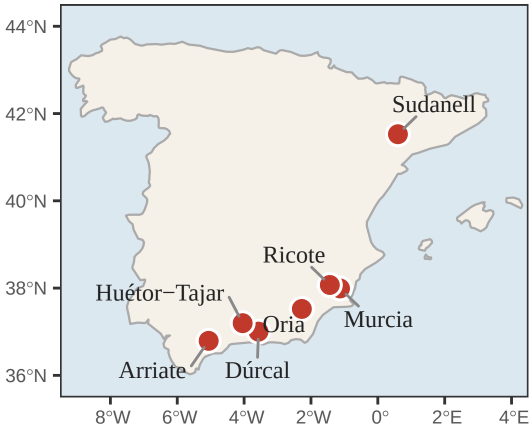
```
# SPAIN BOUNDARY ###############################################################

spain <- ne_countries(scale = "medium", country = "Spain", returnclass = "sf")
```

1.1 Plot map

```
# PLOT #########################################################################

plot_locations <- ggplot() +
  geom_sf(data = spain, fill = "#f5f0e8", color = "#aaaaaa", linewidth = 0.4) +
  geom_point(data = locations,
             aes(x = lon, y = lat),
             shape = 21, fill = "#c0392b", color = "white",
             size = 3.5, stroke = 0.8) +
  geom_text_repel(data = locations,
                  aes(x = lon, y = lat, label = place),
                  size = 3.2, family = "serif", color = "#222222",
                  box.padding = 0.5, point.padding = 0.3,
                  segment.color = "#888888", segment.linewidth = 0.4,
                  min.segment.length = 0.2,
                  max.overlaps = Inf, seed = 42) +
  coord_sf(xlim = c(-9.5, 4.5), ylim = c(35.5, 44.5), expand = FALSE) +
  labs(title = NULL, x = NULL, y = NULL) +
  theme_AP() +
  theme(
    panel.background = element_rect(fill = "#dce8f0", color = NA),
    panel.grid       = element_line(color = "#cccccc", linewidth = 0.2),
    axis.text        = element_text(size = 8, color = "#555555"),
    plot.margin      = margin(10, 10, 10, 10)
  )
```

```
## Warning in geom_text_repel(data = locations, aes(x = lon, y = lat, label =
## place), : Ignoring unknown parameters: `segment.linewidth`
```

```
plot_locations
```


44°N
42°N
40°N
38°N
36°N
8°W
6°W
4°W
2°W
0°
2°E
4°E
Sudanell
Ricote
Huétor−Tajar
Oria
Murcia
Arriate
Dúrcal

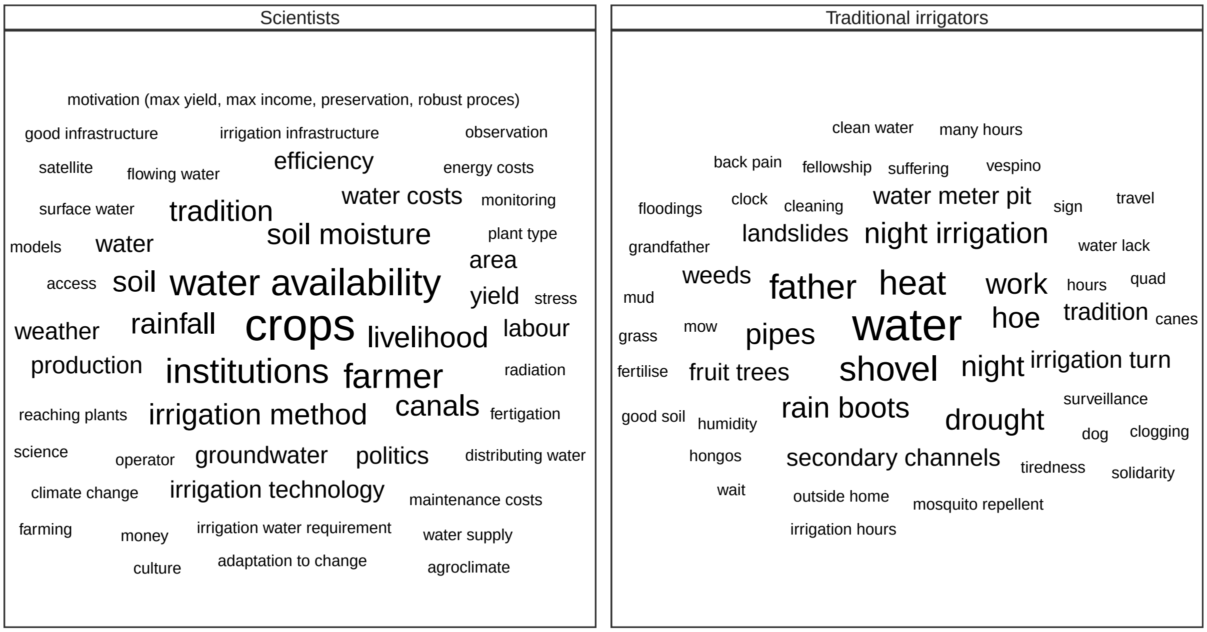
1.2 Load vocabulary data

```r
# LOAD VOCABULARY DATA ########################################################

# Irrigators: factors sheet, general list, English translation ----------------

irrigators <- as.data.table(
  read_xlsx(here("datasets", "input", "carmen_dataset_2.xlsx"), sheet = "factors")
)
irrigators_wc <- irrigators[list == "general" & !is.na(factor_en) & place == "sudanell" ,
                             .(n = .N), by = .(word = factor_en)]
irrigators_wc[, group := "Traditional irrigators"]

# Scientists: Sheet2, general list, homogenized concepts ---------------------
# The raw `factor` column has many near-synonyms and a few typos. We map
# them to canonical concepts via a lookup table (raw -> canonical) so the
# wordclouds (and everything downstream) work on homogenized terms.
# See datasets/input/scientists_general_factor_lookup.csv for the mapping.

scientists <- as.data.table(
  read_xlsx(here("datasets", "input", "scientists_vocabulari.xlsx"), sheet = "Sheet2")
)
factor_lookup <- fread(here("datasets", "input",
                            "scientists_general_factor_lookup.csv"))
scientists[, factor_canonical := homogenize_factors_fun(factor, factor_lookup)]

scientists_wc <- scientists[list == "general" & !is.na(factor_canonical),
                             .(n = .N), by = .(word = factor_canonical)]
scientists_wc[, group := "Scientists"]

# Top 50 per group -----------------------------------------------------------

irrigators_wc <- setorder(irrigators_wc, -n)[1:50]
scientists_wc  <- setorder(scientists_wc,  -n)[1:50]

# Combine ---------------------------------------------------------------------

wc_data <- rbind(irrigators_wc, scientists_wc)
```

1.3 Plot wordcloud

```r
# WORDCLOUD ###################################################################

plot_wordcloud <- ggplot(wc_data, aes(label = word, size = n)) +
  geom_text_wordcloud(seed = 42, color = "black") +
  scale_size_area(max_size = 6) +
  facet_grid(. ~ group) +
```

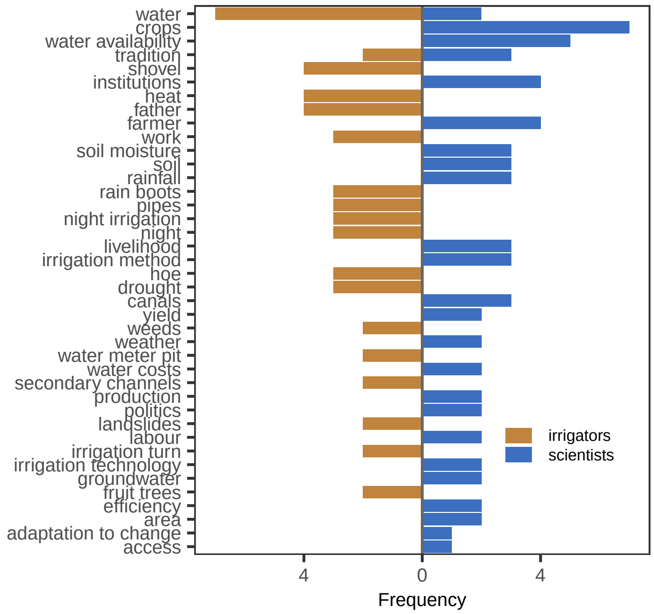
```
  theme_AP()

plot_wordcloud
```



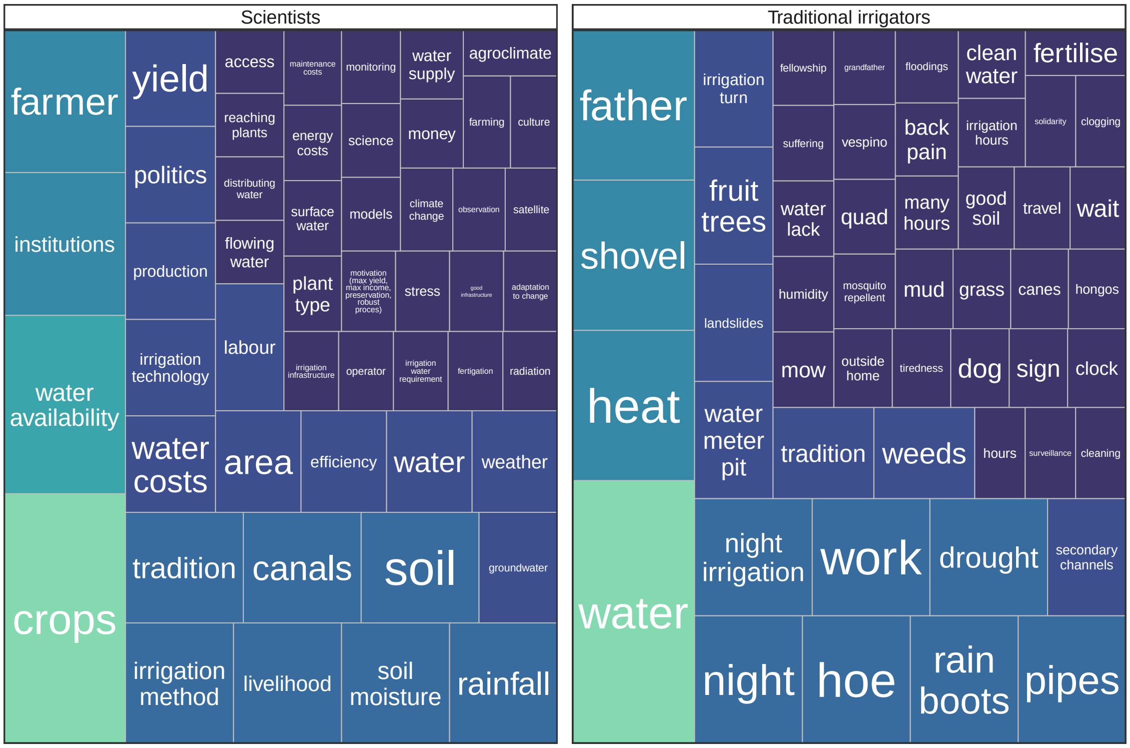
1.4 Pyramid / diverging bar

```
# PLOT PYRAMID / DIVERGING BAR ##################################################

# Words on the y-axis, irrigators counts going left, scientists going right.
# Direct visual comparison: shared vs group-specific terms pop out at once.
wc_wide <- dcast(wc_data, word ~ group, value.var = "n", fill = 0)
setnames(wc_wide, c("Traditional irrigators", "Scientists"),
         c("irrigators", "scientists"))
wc_wide[, total := irrigators + scientists]
wc_pyr <- wc_wide[order(-total)][seq_len(min(.N, 40))]
wc_pyr_long <- melt(wc_pyr, id.vars = "word",
                    measure.vars = c("irrigators", "scientists"),
                    variable.name = "group", value.name = "n")
wc_pyr_long[group == "irrigators", n := -n]

plot_alt_pyramid <- ggplot(wc_pyr_long,
                           aes(x = n,
                               y = reorder(word, abs(n), sum),
                               fill = group)) +
  geom_col() +
  geom_vline(xintercept = 0, color = "grey40") +
  scale_x_continuous(labels = abs) +
  scale_fill_manual(values = c(irrigators = "#c0843d",
                               scientists = "#3d6fc0")) +
```

```r
  labs(x = "Frequency", y = NULL, fill = NULL) +
  theme_AP() +
  theme(legend.position = c(0.8, 0.2))

plot_alt_pyramid
```




```r
# PLOT TREEMAP ################################################################

plot_alt_treemap <- ggplot(wc_data,
                           aes(area = n, label = word, fill = n)) +
  geom_treemap() +
  geom_treemap_text(color = "white", place = "centre",
                    grow = FALSE, reflow = TRUE, min.size = 1) +
  scale_fill_viridis_c(option = "mako", begin = 0.25, end = 0.85,
                       guide = "none") +
  facet_wrap(~ group) +
  theme_AP()

plot_alt_treemap
```


Scientists
farmer
institutions
water availability
crops
yield
politics
production
irrigation technology
water costs
access
reaching plants
distributing water
flowing water
labour
maintenance costs
energy costs
surface water
plant type
monitoring
science
models
water supply
money
agroclimate
farming
culture
climate change
observation
satellite
motivation (max yield, max income, preservation, robust proces)
stress
good infrastructure
adaptation to change
irrigation infrastructure
operator
irrigation water requirement
fertigation
radiation
area
efficiency
water
weather
tradition
canals
soil
groundwater
irrigation method
livelihood
soil moisture
rainfall
Traditional irrigators
father
shovel
heat
water
irrigation turn
fruit trees
landslides
water meter pit
fellowship
grandfather
floodings
clean water
fertilise
suffering
vespino
back pain
irrigation hours
solidarity
clogging
water lack
quad
many hours
good soil
travel
wait
humidity
mosquito repellent
mud
grass
canes
hongos
mow
outside home
tiredness
dog
sign
clock
tradition
weeds
hours
surveillance
cleaning
night irrigation
work
drought
secondary channels
night
hoe
rain boots
pipes

1.5 Calculation of Spain's share

We use the ten global irrigated-area datasets compiled by Puy et al. (2026): GAEZ+2015, GIAM, GMIA, GRIPC, LUH2, Meier, MIRCA 2000, MIRCA OS, Nagaraj, and SPAM2010. For each dataset and spatial resolution (0.2°, 0.4°, 1°), we compute Spain's irrigated area (Mha), total European irrigated area (Mha), and Spain's percentage share.

```r
# READ IN DATASETS ##############################################################

country_level_irrigated_areas <- fread(here("datasets", "input",
                                            "country_level_irrigated_areas.csv"))



# Calculate ---------------------------------------------------------------------

country_level_irrigated_areas[continent == "Europe",
                              .(spain_mha = sum(mha[country == "Spain"]),
                                europe_mha = sum(mha),
                                spain_share_pct = 100 * sum(mha[country == "Spain"]) /
                                  sum(mha)),
                              by = .(dataset, resolution)]
```

```
##       dataset resolution spain_mha europe_mha spain_share_pct
##        <char>     <char>     <num>      <num>           <num>
##  1:   gaez_v4     0.2deg  3.115963   23.84897       13.065402
##  2:      giam     0.2deg  2.702138   38.65570        6.990269
##  3:      gmia     0.2deg  3.231380   11.95049       27.039725
##  4:     gripc     0.2deg  4.003731   23.03500       17.381072
##  5:      luh2     0.2deg  3.135771   13.78261       22.751647
##  6:     meier     0.2deg  4.223571   26.00185       16.243345
##  7: mirca2000     0.2deg  3.063020   15.32320       19.989427
##  8:  mirca_os     0.2deg  2.872767   10.92628       26.292260
##  9:   nagaraj     0.2deg  4.651206   32.00352       14.533421
## 10:      spam     0.2deg  2.723286   10.92166       24.934715
## 11:   gaez_v4     0.4deg  3.088649   23.62507       13.073608
## 12:      giam     0.4deg  2.686260   38.53872        6.970287
## 13:      gmia     0.4deg  3.192415   11.81225       27.026305
## 14:     gripc     0.4deg  3.966339   23.03867       17.216008
## 15:      luh2     0.4deg  3.123777   13.73316       22.746234
## 16:     meier     0.4deg  4.177631   25.64511       16.290168
## 17: mirca2000     0.4deg  3.043205   15.29263       19.899813
## 18:  mirca_os     0.4deg  2.840007   10.87587       26.112921
## 19:   nagaraj     0.4deg  4.601698   31.51188       14.603057
## 20:      spam     0.4deg  2.692517   10.80322       24.923284
## 21:   gaez_v4       1deg  2.936647   23.21744       12.648451
## 22:      giam       1deg  2.558272   38.42329        6.658128
## 23:      gmia       1deg  3.056154   11.54011       26.482897
## 24:     gripc       1deg  3.745961   22.80496       16.426079
## 25:      luh2       1deg  2.948508   13.17117       22.386067
```

```
## 26:     meier       1deg  3.928146   25.11399        15.641264
## 27: mirca2000       1deg  2.887533   14.99172        19.260849
## 28:  mirca_os       1deg  2.712319   10.69960        25.349720
## 29:   nagaraj       1deg  4.272359   30.64785        13.940160
## 30:      spam       1deg  2.559707   10.50107        24.375677
##       dataset resolution spain_mha europe_mha spain_share_pct
##        <char>     <char>     <num>      <num>           <num>
```

2 References

Puy, Arnald, Olivia Richards, Seth Nathaniel Linga, Samuel Flinders, Carmen Aguiló-Rivera, and Josh Larsen. 2026. "Code and Datasets of Where Irrigation Exists Is Globally Contested." *Zenodo*, ahead of print, March. https://doi.org/10.5281/zenodo.19001232.

3 Session information

```
# SESSION INFORMATION ###########################################################

sessionInfo()
```

```
## R version 4.5.2 (2025-10-31)
## Platform: aarch64-apple-darwin20
## Running under: macOS Tahoe 26.3.1
##
## Matrix products: default
## BLAS:   /System/Library/Frameworks/Accelerate.framework/Versions/A/Frameworks/vecLib.framew
## LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib;
##
## locale:
## [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
##
## time zone: Europe/London
## tzcode source: internal
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
##  [1] treemapify_2.5.6        benchmarkme_1.0.8       cowplot_1.2.0.9000
##  [4] ggwordcloud_0.6.2       readxl_1.4.5            here_1.0.2
##  [7] ggrepel_0.9.6           rnaturalearthdata_1.0.0 rnaturalearth_1.1.0
## [10] sf_1.1-0                ggplot2_4.0.3.9000      data.table_1.18.2.1
## [13] sensobol_1.1.9
##
## loaded via a namespace (and not attached):
##  [1] ggfittext_0.10.2     benchmarkmeData_1.0.4 gtable_0.3.6
##  [4] xfun_0.55            lattice_0.22-7        vctrs_0.7.3
##  [7] tools_4.5.2          Rdpack_2.6.6          generics_0.1.4
## [10] parallel_4.5.2       tibble_3.3.1          proxy_0.4-29
## [13] pkgconfig_2.0.3      Matrix_1.7-4          KernSmooth_2.23-26
## [16] RColorBrewer_1.1-3   S7_0.2.2              lifecycle_1.0.5
## [19] stringr_1.6.0        compiler_4.5.2        farver_2.1.2
## [22] tinytex_0.58         codetools_0.2-20      litedown_0.7
## [25] htmltools_0.5.9      class_7.3-23          yaml_2.3.12
## [28] pillar_1.11.1        classInt_0.4-11       iterators_1.0.14
## [31] foreach_1.5.2        commonmark_2.0.0      tidyselect_1.2.1
## [34] digest_0.6.39        stringi_1.8.7         dplyr_1.1.4
## [37] labeling_0.4.3       rprojroot_2.1.1       fastmap_1.2.0
## [40] grid_4.5.2           colorspace_2.1-2      cli_3.6.6
## [43] magrittr_2.0.5       e1071_1.7-17          withr_3.0.2
## [46] scales_1.4.0         rmarkdown_2.30        httr_1.4.8
## [49] otel_0.2.0           cellranger_1.1.0      png_0.1-8
```

```
## [52] evaluate_1.0.5        knitr_1.51            rbibutils_2.4.1
## [55] doParallel_1.0.17     viridisLite_0.4.3     markdown_2.0
## [58] rlang_1.2.0           gridtext_0.1.5        Rcpp_1.1.1-1.1
## [61] glue_1.8.1            DBI_1.3.0             xml2_1.5.2
## [64] rstudioapi_0.17.1     R6_2.6.1              units_1.0-0
```

```r
## Return the machine CPU ----------------------------------------------------

cat("Machine:     "); print(get_cpu()$model_name)
```

```
## Machine:
## [1] "Apple M1 Max"
```

```r
## Return number of true cores -------------------------------------------------

cat("Num cores:   "); print(parallel::detectCores(logical = FALSE))
```

```
## Num cores:
## [1] 10
```

```r
## Return number of threads ----------------------------------------------------

cat("Num threads: "); print(parallel::detectCores(logical = FALSE))
```

```
## Num threads:
## [1] 10
```